

Large language model based system with causal inference and Chain-of-Thoughts reasoning for traffic scene risk assessment

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ABSTRACT

Evaluating potential traffic scene risks is crucial for the decision-making process in autonomous driving systems. Large Language Models (LLMs), due to their advanced scene understanding and reasoning capabilities, offer a new paradigm for decision-making in autonomous driving. However, when used to evaluate traffic scene risks, LLM faces challenges of model hallucination and slow inference speed. Therefore, we have developed a TrafficRiskGPT-based system. Initially, we designed the TrafficRiskGPT model, which is a large language model specifically used for traffic scene risk reasoning. Its core is based on the LLaMA3-8B model, and it incorporates a large amount of traffic risk datasets for LoRA (Low-Rank Adaptation) fine-tuning, enabling the model to deeply understand traffic scene risks. On this basis, we designed a system around the TrafficRiskGPT model. The system establishes a knowledge base based on traffic scene risks and incorporated HNSW (Hierarchical Navigable Small World graphs) method to improve the retrieval efficiency of the knowledge base, introduces vLLM technology to improve the system's inference speed, and constructs a comprehensive risk evaluation metric to assess the system's performance on traffic scene risks. Finally, we designed a CI-CoT (Causal Inference Chain-of-Thought) technique, allowing the system to gradually evaluate the traffic risks associated with each decision, thereby reducing model hallucinations and slow inference speed issues. Our experiments show that in the same scenarios, in terms of vehicle collision rates, our method reduces the rate by 7.3% compared to GPT4o-mini, significantly reduces outputs irrelevant to traffic scene risks, and improves model inference speed by a factor of 3.62.

1. Introduction

Traffic scene risk assessment is a crucial means of ensuring vehicle traffic safety. The research problem of traffic scene risk assessment is how to accurately analyze the potential risks in the surrounding traffic scene using the collected traffic data in order to cope with the challenges brought by the dynamic interaction of multiple traffic participants. Traditional methods for conducting traffic risk assessments include the Analytic Hierarchy Process (AHP) [1], Bayesian theory [2], and rule-based approaches [3]. These methods mainly handle structured data, making it difficult to fully utilize unstructured data sources such as natural language descriptions. In addition, they lack a deep understanding of the context of traffic scene risks, which may lead to inaccuracies in traffic scene risk assessments. Recently, large language models (LLMs), with their robust capabilities in language understanding and reasoning, have introduced a new paradigm to the field of autonomous driving and human-robot interaction [4]. This has also

brought new opportunities to the domain of traffic risk assessment. In the field of autonomous driving, a multitude of studies have introduced diverse frameworks for LLM-based autonomous driving, including Lim-Sim++ [5], DiLu [6], and DriveLLM [7], which enhance autonomous driving through LLM's superior scene comprehension abilities and employ simulation platforms such as HighwayEnv [8] and CARLA [9] to ascertain the frameworks' efficacy. Upon analyzing these frameworks, we have delineated the foundational paradigm of LLM-based autonomous driving, illustrated in Fig. 1, comprising four components: environment, facility, LLM and agent. These four components form a closed loop and continuously iterate to achieve LLM-based autonomous driving. Our research motivation is to integrate the powerful reasoning capabilities of LLMs to overcome the limitations of traditional methods, such as poor adaptability and generalization ability, thereby improving traffic scene assessment and enhancing traffic safety.

However, when used to evaluate complex traffic scene risks, LLM [10,11] faces challenges of model hallucinations and slow inference

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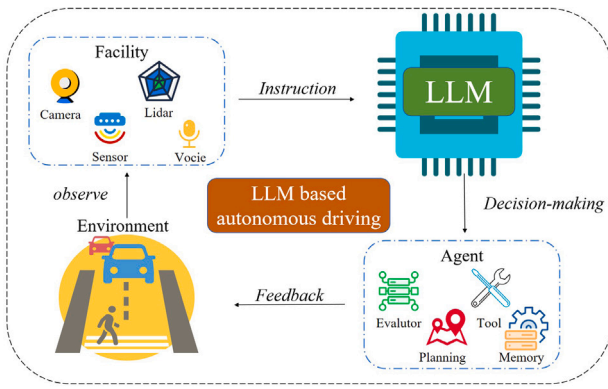


Fig. 1. The basic paradigm of autonomous driving based on LLM. We have conducted relevant summaries, it generally consists of four parts: environment, facility, LLM, and agent.

speed. On the one hand, the model illusions may mislead driving decisions and increase traffic safety risks. On the other hand, due to the constrained computational and storage resources of onboard computers or intelligent edge devices, large models such as GPT-3.5, with a parameter count of up to 175 billion, are not directly deployable on vehicles. The intricate structure and substantial computational burden of LLMs present significant hurdles to real-time decision-making. One solution is to use APIs to interact with LLMs, but waiting for API response would lead to delays in vehicle decision-making, particularly in high-speed scenes characterized by rapidly changing circumstances [12,13]. Such delays not only elevate the risk of traffic accidents but also potentially endanger vehicle passengers and other road users. Thus, tackling these challenges necessitates a series of interventions. For unreliable content generated by models, a local knowledge base can be constructed to alleviate hallucinations. For onboard devices constrained by resources, the reduction of computational burden can be attained by optimizing LLM structures, compressing parameters, and accelerating LLMs. To meet the demands of real-time decision-making, exploring the deployment of compressed LLMs locally can enable faster inference and response. Furthermore, the adoption of a hybrid intelligent system, which integrates decision-making based on LLMs with conventional knowledge bases, can bolster decision stability and reliability.

In this paper, we design a large language model specifically for traffic scene risk reasoning, called the TrafficRiskGPT model. Its core is based on the LLaMA3-8B model, and it incorporates a large amount of traffic risk datasets for LoRA (Low-Rank Adaptation) fine-tuning, enabling the model to deeply understand traffic scene risks. Based on the TrafficRiskGPT model, we build a system that employs a knowledge base based on traffic scene risks and incorporated HNSW (Hierarchical Navigable Small World Graphs) method to improve the retrieval efficiency of the knowledge base, introduces vLLM technology into the system to accelerate the inference process, and uses indicators such as TTC (Time to Collision), DRAC (Deceleration Rate to Avoid a Crash), and DSS (Difference between Space Distance and Stopping Distance) to evaluate the effectiveness of the system. We also design a CI-CoT (Causal Inference Chain-of-Thought) technique to enhance the model's interpretability and decision reliability. Finally, our method reduces the vehicle collision rate by 64.43% compared to ChatGPT 3.5, significantly reduces outputs irrelevant to traffic scene risks, and improves the model inference speed by 3.62 times. This indicates that our system effectively addresses the issues of model illusion and inference speed, providing insights for using large-scale language models for traffic risk assessment.

In summation, our contributions are delineated as follows:

- We elaborately design a traffic scene risk reasoning system based on LLM, focusing on comprehensive and in-depth understanding of traffic risk scenes.

- We propose a customized LLM for traffic scene risk reasoning, TrafficRiskGPT, based on the LLaMA3-8B architecture and fine-tuned with LoRA using a vast amount of traffic risk datasets. This enables the model to understand traffic scene risks deeply.
- We construct a knowledge base centered on traffic scene risks to reduce model hallucinations. Additionally, we incorporated the HNSW method to improve the retrieval efficiency of the knowledge base, and we introduced vLLM technology to significantly enhance the model's reasoning speed and designed a comprehensive traffic risk evaluation index to assess the model's performance thoroughly.
- We develop a CI-CoT technique that captures potential traffic scene risk factors at each step of the reasoning process. This approach avoids generating information inconsistent with actual scenarios and improves the model's reliability.

The remaining part of this article is structured as follows. In Section 2, we elaborated on the relevant work in detail. In Section 3, we introduced the overall system of TrafficRiskGPT. In section, we introduced the relevant technologies of the TrafficRiskGPT system. In Section 5, we conducted relevant experimental simulation tests. Section 6 summarizes the entire text.

2. Related work

In this section, we describe the related work from the aspects of LLM-based autonomous driving applications, traffic scene risk assessment and LLM inference.

2.1. LLM-driven framework

Due to its excellent performance in tasks such as understanding and inference of the scene, LLM has sparked a series of new paradigms for autonomous driving based on LLM. Fu et al. [5] developed an improved LimSim++ platform specifically tailored for using LLMs in autonomous vehicle technology. This platform provides long-term and multi-scenario simulation, providing key information for multimodal LLM driven vehicles. Wen et al. [6] proposed the DiLu framework to address challenges such as dataset bias, overfitting, and inexplicability. This framework combines reasoning and reflection modules, enabling the system to make decisions based on common sense and continuously evolve. Wang et al. [14] introduced the LLM-based autonomous driving framework DriveMLM, which can perform closed-loop autonomous driving in real simulators, exploring new opportunities for intelligent agents by granting them capabilities akin to human thought and cognition. Tanahashi et al. [15] conducted quantitative research on how different types of LLMs accurately handle the spatial perception decision-making ability and traffic rule compliance ability of autonomous driving, and developed a system for driving vehicles using LLMs. Cui et al. [16] proposed a new framework that utilizes LLMs to improve decision-making in autonomous vehicles. Sha et al. [17] used LLM as decision-making elements in complex autonomous driving scenarios that demand an understanding of human common sense. Xu et al. [18] launched a new interpretable end-to-end auto drive system DriveGPT4 based on LLM. DriveGPT4 can handle multiple frames of video input and text queries, helping to interpret vehicle actions, provide relevant reasoning, and effectively solve various problems raised by users. LLMs have made significant progress in integrating with autonomous driving systems, but relatively few studies have focused on methods related to traffic scene risk. Recognizing this, our work specifically constructs an LLM that serves traffic scene risk, enabling autonomous driving systems to pay more attention to traffic scene risk, thereby enhancing traffic safety.

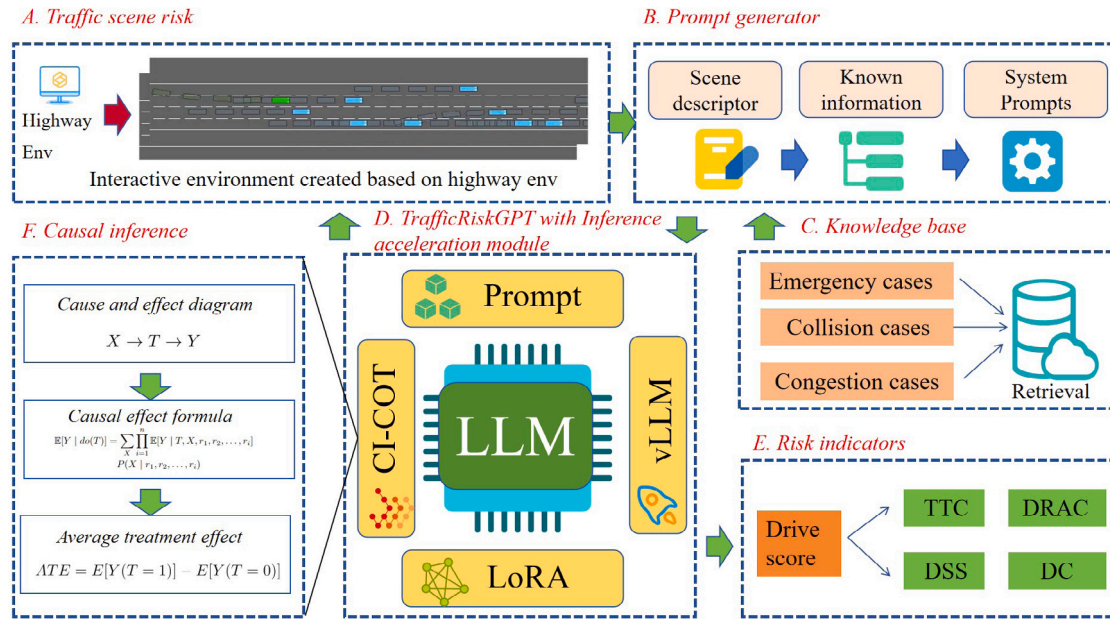


Fig. 2. The system of TrafficRiskGPT. This system consists of technical methods such as prompt generator, knowledge base, TrafficRiskGPT model, etc.

2.2. Traffic scene risk assessment

Quantifying traffic risks and conducting assessments [19,20] is a key direction of research for many scholars. Gao et al. [21] proposed a scene assessment method based on uncertainty risk analysis, which combines Gaussian distribution model and random model to evaluate collision probability through trajectory prediction. This method can conduct traffic scenario risk assessment in the case of sensor failure and communication packet loss. Li et al. [22] proposed a probabilistic model-based traffic risk assessment method that combines conditional random fields to evaluate the risk level of surrounding scenes. This method can avoid collisions in multiple scenes with different driving style preferences. Huang et al. [23] proposed a driving risk assessment model. Firstly, a driving risk coupling module was constructed, which combines the theoretical concepts of internal and external fields. Then, a polynomial model was used to calculate the traffic risk coefficient and compare it with the TTC risk assessment index. The proposed method is effective in assessing driving risk. Liu et al. [24] proposed a pedestrian collision risk assessment model based on Transformer and GCN to evaluate pedestrian collision risk, and constructed a traffic scene graph dataset for model validation, and published relevant source code and dataset. Yang et al. [25] designed a multi-level assessment system for intersection risk. The system first quantifies traffic risk based on the frequency and severity of traffic conflicts, and then models the nonlinear relationship between traffic conflict points using the plume diffusion model, flexibly assessing intersection risk in a fine-grained manner. Although previous research has made significant progress in traffic risk assessment, relatively few studies have used traffic risk assessment methods to evaluate the decision-making capabilities of LLM models in traffic scenes. Traditional risk assessment methods often rely on manually designed rules and models, which are difficult to fully address the dynamic complexity in traffic scenes. LLM can learn a large amount of general knowledge and traffic risk knowledge through pre-training, and can also be fine-tuning by combining specific traffic data to further enhance its adaptability to specific traffic risk scenes.

2.3. LLMs for reasoning

Recent research has greatly improved the reasoning ability of LLM by drawing on methods such as thought chains and knowledge graphs.

Chang et al. [26] introduced SocraSynth, a multi LLM proxy inference platform aimed at reducing model hallucinations and improving inference ability. Wang et al. [27] tested LLM's reasoning ability by engaging in debate like conversations with LLM and subjected them to rigorous testing across a range of complex reasoning tasks, including mathematics, common sense, logic, and BIG Bench challenges. Hsu et al. [28] suggested a Symbol-LLM system designed to derive visual content from images and interpret it as symbols and infer activity semantics based on rules through fuzzy logic calculation. Luo et al. [29] proposed a planning retrieval reasoning framework that can collaborate LLM with knowledge graphs to achieve faithful and interpretable reasoning. Wang et al. [30] proposed a new method for constructing personalized inference graphs using LLMs, which is composed of four elements: inference via chain graphs, expansive extensions, self-assessment with scoring, and the self-enhancement of the knowledge base to increase the efficacy of recommendations. Zhang et al. [31] proposed a new technique known as Solver Layer Adaptation (SoLA) that incorporates a solver into the architecture of large language models as an additional layer. This approach directs the model towards achieving solutions that satisfy varying conditions. Our analysis of SoLA across several datasets empirically establishes its superior efficacy, outshining traditional symbol solvers like Z3 and Kissat as well as conventional machine learning approaches in resolving complex, large-scale issues. Previous research has made many contributions in LLM inference, but these methods may face slow inference speeds and lack the ability to capture details of traffic risk factors during the inference process. Our work builds a knowledge base based on traffic scene risk, while combining vLLM and CI-COT technologies to improve the inference speed of the TrafficRiskGPT model and its ability to discover causality in traffic scene risks.

3. TrafficRiskGPT-based system

We propose an LLM-based system with a knowledge base inference acceleration module, which promotes the attention of autonomous driving systems to traffic scene risks, thereby enhancing driving safety. The overall framework is illustrated in Fig. 2. The core of the system is our proposed TrafficRiskGPT, which is an LLM based on the LLaMa3-8B model fine-tuned through LoRA for traffic risk. First, we use a scene descriptor to transform the surrounding traffic scene of vehicles into

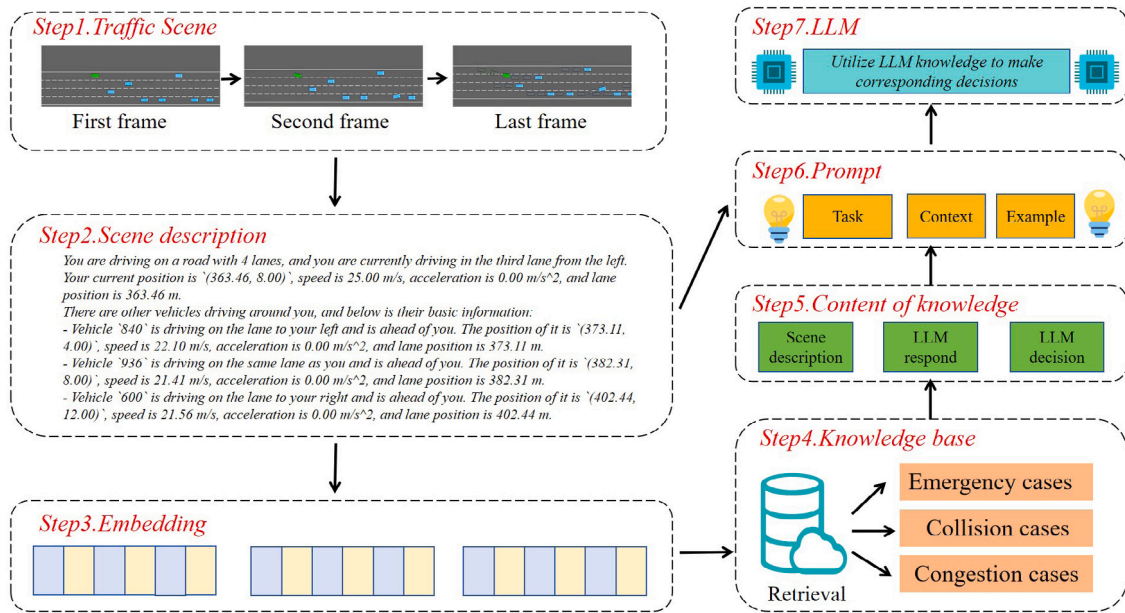


Fig. 3. Building a knowledge base. The main focus is to convert video data into scenes described in natural language.

natural language text, combined with some known information such as traffic regulations, system prompts like collision warnings, and integrated multiple traffic scene cases (including emergency cases, collision cases, congestion cases, etc.) in the knowledge base of the vehicle side. Through structured processing, the prompt information is formed and input into the TrafficRiskGPT system. Then, to facilitate deployment on in-vehicle platforms, vLLM is introduced to reduce the computational resource consumption of TrafficRiskGPT. To better assess the effectiveness of the proposed TrafficRiskGPT, a driving evaluation score index is constructed by integrating TTC, DRAC, DSS, and DC. Finally, we introduce causal inference combined with COT technology to further enhance the reasoning ability and interpretability of TrafficRiskGPT. The details of the system are as follows.

Traffic scene risk. We use HighwayEnv software to simulate traffic risk scenarios. Through this software, we can construct a variety of complex traffic scenes to simulate real-world road traffic conditions. In the simulation process, we can set various traffic risk factors, such as weather conditions, traffic accidents, etc., to study the impact of various factors on traffic safety. Through simulation, we can optimize road traffic conditions, improve traffic flow allocation, and reduce congestion and accidents. Meanwhile, the simulation results provide valuable data support for LLM, helping it make more scientific decisions.

Prompt generator. The generation of prompt words involves three core components: the scene descriptor, known information, and system prompts. The scene descriptor's role is to convert complex and variable traffic scenes into easily understandable natural language descriptions, ensuring that, even in highly dynamic environments, the system can accurately capture key information. Known information refers to traffic regulations and other relevant conventional information that must be followed, such as speed limits and interpretations of traffic signs, providing a basis for the system's code of conduct. System prompts include collision warning information, etc.

Knowledge base. To enhance the reasoning ability of LLMs in handling complex closed-loop driving tasks, we constructed a case dataset specifically targeted at the domain of traffic risk and built a local traffic risk knowledge base on the foundation of this dataset. As shown in Fig. 3, we first convert the scenes from the dataset into natural language text forms and then transform the text into vector representations in high-dimensional space through embedding. Based on this, we define a function that transforms scene descriptions, LLM

responses, decision-making knowledge, and questions into vectors in high-dimensional space for efficient retrieval and reasoning.

For a given input x , its embedded representation is $E(x)$. The knowledge base can be represented as a set of embedded vectors.

$$KB = \{E(x_1), E(x_2), \dots, E(x_n)\}, \quad (1)$$

where x_i represents an element in the knowledge base, such as a scene description, an LLM response, or a piece of decision-making knowledge.

On this basis, we introduce an HNSW (Hierarchical Navigable Small World graphs) retrieval method to further enhance the efficiency of knowledge retrieval. This method uses a layered graph structure, with each layer being a small-world graph. HNSW mainly consists of two stages: one is the construction stage, and the other is the search stage.

In the construction stage, the HNSW method samples each data point according to a certain probability distribution and constructs the NSW graphs [32] layer by layer.

In the search stage, the most critical step is to find the neighborhood that is most relevant to the query point. The algorithm starts from the top layer and searches for candidate nodes while gradually narrowing down the search radius layer by layer until the lowest layer is reached. At this point, it continues to search in a smaller radius until the nearest neighbors are finally collected. In each search update step, the HNSW method calculates the distance and chooses the next node set for updating, which is:

$$D_{\text{dis}} = \min\{D_{\text{dis}}(q, p_i), D_{\text{dis}}(q, p_j)\}, \quad (2)$$

where q is the query point, and p_i and p_j are candidate nodes. $D_{\text{dis}}(\bullet)$ represents the distance function.

The core idea of HNSW is to achieve high-efficiency nearest neighbor search by constructing a sparse graph, thereby enabling fast knowledge retrieval.

TrafficRiskGPT with an inference acceleration module. TrafficRiskGPT with the inference acceleration module is a key technical component of the traffic risk framework. This module is designed to enable the in-vehicle system to efficiently analyze the surrounding traffic scenes, quickly identify potential risks, and, combined with the inference acceleration module, complete complex risk analysis tasks in a short period, thereby making accurate traffic risk decisions. In Section 4, we will delve into the technical details and implementation methods of this module.

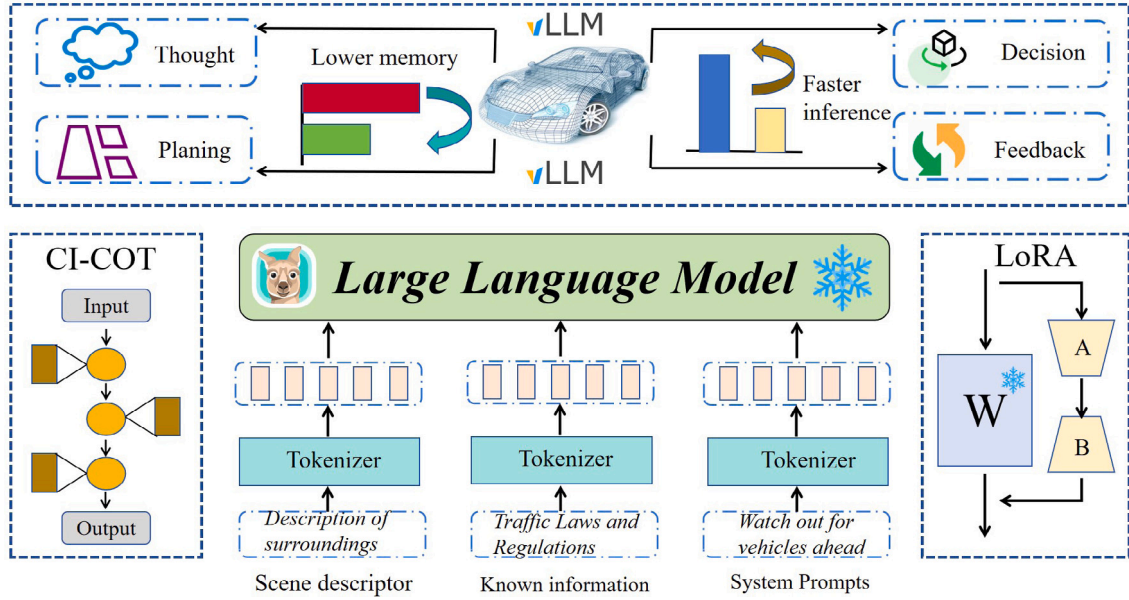


Fig. 4. The basic model of TrafficRiskGPT.

Risk indicators. We employ a multi-dimensional set of risk indicators, which includes TTC, DRAC, DSS, and DC. These indicators essentially cover various road traffic scenarios. In Section 4, we will delve into the evaluation metrics of this module.

CI-CoT. This section introduces the technical method based on CI-CoT, which enables the TrafficRiskGPT model to consider traffic causality during the inference process, which we will provide a detailed description of in Section 4.

4. TrafficRiskGPT with inference acceleration module related technologies

In this section, we describe the designed TrafficRiskGPT and its related technologies. It includes how to use LoRA to fine-tune LLM, related technologies of vLLM, traffic risk assessment indicators, and CI-CoT.

4.1. TrafficRiskGPT

We have built a LLM based on traffic risk, TrafficRiskGPT, as shown in Fig. 4. We generate prompt words based on the scene descriptor, known information, and system prompts, and then convert them into tokens to be imported into the LLM. We chose LLaMa3-8B as the base model [33], a high-performance large language model with strong capabilities in understanding and generating natural language. To further optimize this model to suit complex traffic risk assessment scenes, we employed Low-Rank Adaptation (LoRA) technology for fine-tuning [34]. LoRA adjusts the weight matrices of the transformer's layers. For a given weight matrix $W \in \mathbb{R}^{d \times k}$, the adjustment made by LoRA can be represented as:

$$W' = W + \Delta W = W + PQ, \quad (3)$$

where W' is the adjusted weight matrix, ΔW is the low-rank modification, $P \in \mathbb{R}^{d \times r}$, and $Q \in \mathbb{R}^{r \times k}$. This maintains a balance between adaptability and model complexity.

During the training process of the traffic risk large language model, it is necessary to freeze the parameter W and only train the parameters in P and Q . Therefore, the forward propagation process becomes:

$$h = Wx + \Delta Wx = Wx + PQx. \quad (4)$$

During the initialization process, we have:

$$h = W_0x + \frac{\alpha}{r} PQx, \quad (5)$$

where W_0 is the old knowledge in pre-training weights, $\frac{\alpha}{r} PQx$ is the approximation of incremental weights ΔW .

Through LoRA technology, we have successfully optimized the LLaMa-7B model to make it more focused on risk analysis and decision-making in traffic scenes. This has not only increased the model's specialization and precision but also significantly improved the efficiency and effectiveness of the model in handling traffic risk assessment tasks.

4.2. vLLM inference acceleration module

On the vehicle side, limited by hardware resources such as processor speed, memory size, etc, there is often a situation of strained computational resources. It is very necessary to make reasonable use of resources when deploying large models in vehicles. To enhance the inference speed of large models while reducing their memory consumption, we introduced the vLLM acceleration platform [35], which is designed specifically to optimize the running efficiency of LLM in resource-constrained environments.

vLLM mainly has the following characteristics: the first is high service throughput, the second is memory management of attention keys and values through PagedAttention to reduce computational resource consumption, the third is batch processing of incoming requests, and the fourth is optimization for the CUDA kernel. Based on the above characteristics, deploying vLLM technology to the vehicle can significantly reduce computational resource overhead and improve the speed of traffic risk language models.

Although vLLM reduces resource consumption, TrafficRiskGPT still has significant computational requirements. According to [36,37], we can combine edge cloud collaboration to alleviate the enormous computing pressure that TrafficRiskGPT brings to the in-vehicle system, especially in scenarios where resources are limited or high-frequency real-time inference is required. The in-vehicle system can first perform preliminary inference processing, and for some complex computing tasks, they can be forwarded to the cloud for processing. This can maintain real-time performance and reduce the burden on the in-vehicle computing devices. In addition, edge computing devices can

also dynamically adjust the computing load according to the complexity of the task to ensure the best performance under limited resources.

We referred to the method in Ref. [36] and proposed a cloud based collaborative inference system [38] based on dynamic programming algorithm to minimize the latency of TrafficRiskGPT model inference and maximize the throughput of TrafficRiskGPT model. The specific steps are analyzed in detail in Ref. [36] and will not be repeated here. In the experimental section, we will analyze in detail the latency and throughput of the TrafficRiskGPT model under different conditions.

4.3. Traffic risk evaluation model

We have created a module to evaluate traffic risks in autonomous driving LLM, with a focus on safety and comfort. The overall effectiveness will be determined based on the specific generated content to determine whether the model is more accurate in identifying traffic risks. The Risk Score mainly includes risk assessment indicators such as TTC, DRAC, and DSS. DriveScore (DS) not only includes the evaluation of risk indicators, but also considers driving comfort. It is a weighted cumulative comprehensive score of TTC, DRAC, DSS, and DC. The formula for DS is as follows:

$$DS = W_T \cdot TTC_s + W_{DR} \cdot DRAC_s + W_{DS} \cdot DSS_s + W_{DC} \cdot DC_s, \quad (6)$$

where the weights W_T , W_{DR} , W_{DS} and W_{DC} are weight coefficients, which are set differently according to different traffic scenes.

(1) TTC [39]. The level of risk is typically assessed through TTC. The calculation for TTC is as follows:

$$TTC = \frac{x_{n-1}(t) - x_n(t) - b_{n-1}}{v_n(t) - v_{n-1}(t)}, \quad (7)$$

$$TTC_s = \begin{cases} 100, & \text{if } TTC_e \geq TTC_h \\ 100 * \frac{TTC_e}{TTC_h}, & \text{else,} \end{cases} \quad (8)$$

where TTC is the statistical distribution that models the scene in which the rear vehicle n TTC is the statistical distribution that models the scene where the rear vehicle $n - 1$,

$v_{n-1}(t)$ denotes the velocity of the leading vehicle $n - 1$ at time t , while $v_n(t)$ indicates the velocity of the following vehicle n at the same moment, b_{n-1} is the designated safe distance for vehicle $n - 1$, where $v_n(t) > v_{n-1}(t)$. TTC_e refers to the TTC of the ego vehicle, and TTC_h is the threshold for TTC.

(2) DRAC [40]. DRAC means that deceleration rate to avoid collisions. The formula for its evaluation index is:

$$DRAC_s = 100 - \left[\frac{(v_n(t) - v_{n-1}(t))^2}{x_{n-1} - x_n - b_{n-1}} \right] * 100, \quad (9)$$

where $v_n(t) > v_{n-1}(t)$, D_{DRAC} measures the relative risk. A higher D_{DRAC} value indicates a greater risk of collision.

(3) DSS [41]. DSS is determined by the disparity between the space and parking distance, and its formula is as follows:

$$DSS = \frac{v_{n-1}(t)^2}{2ug} + d_n - \left(\frac{v_n(t)\Delta t_n^2}{2ug} \right), \quad (10)$$

$$DSS_s = \begin{cases} 100, & \text{if } DSS_e \geq DSS_h \\ 100 * \frac{DSS_e}{DSS_h}, & \text{else,} \end{cases} \quad (11)$$

where u represents the coefficient of friction; g represents the acceleration due to gravity; d_n is the distance for vehicle n to stop safely; Δt represents the time interval.

(4) DC [42]. The following is the driving comfort model that includes these factors:

$$DC = \left(1 - \frac{|a|}{a_{\max}} \right) \cdot \left(1 - \frac{v_{\text{rel}}}{v_{\max}} \right) \cdot \left(1 - \frac{d_{\min}}{d} \right), \quad (12)$$

$$DC_s = DC * 100, \quad (13)$$

where DC represents the comfort level, ranging between 0 and 1, where 0 indicates extreme discomfort and 1 indicates comfort. v_{rel} represents the relative velocity (m/s) of the vehicle, defined as the speed differential between the vehicle and the object towards which it is moving.

4.4. CI-CoT

To reduce the phenomenon of hallucinations in large language models [43] during traffic decision-making and to improve decision accuracy and interpretability, we introduced the CI-CoT technique. This technique evaluates the traffic risk associated with each action and generates a series of logical steps, ultimately leading to well-reasoned decisions. By doing so, it enhances driving safety and the interpretability of large models. The method generates a series of reasoning steps that allow the model to display its thought process before making the final decision.

For example, given a traffic risk related problem q , TrafficRiskGPT will generate a series of reasoning steps $r_1, r_2, \dots, r_n$ to guide decision d . This can be formulated mathematically as [44]:

$$p(d|r_n) \cdot \prod_{i=1}^n p(r_i|r_{i-1}, \dots, r_1, q), \quad (14)$$

where $p(r_i|r_{i-1}, \dots, r_1, q)$ is the conditional probability of generating reasoning step r_i given a traffic risk related problem q and all previous reasoning steps. The final decision d is made based on the last reasoning step r_n .

We introduce causal inference into Eq. (13), and the equation becomes:

$$p(d|do(r_n)) \cdot \prod_{i=1}^n p(r_i|do(r_{i-1}, \dots, r_1, q)), \quad (15)$$

where $do(r_n)$ represents an intervention on reasoning step r_n , enabling the reasoning of the LLM to better align with real-world traffic scenes, and to place more emphasis on the impact of different causal factors on traffic risk.

In the causal inference model, the model must first identify the following three types of variables: the first type is treatment variables T , including some driving decisions, such as acceleration, deceleration, lane changes, etc. The second type is outcome variables Y , which include certain traffic risk indicators, such as collision risk, driving comfort level, etc. The third type is covariates X , including ego car states (position, speed, acceleration, etc.) and surrounding vehicle information (position, speed, acceleration, etc.).

In traffic scene, we need to establish a causal graph to clarify the causal relationships between variables, so that we can use causal inference to analyze the impact of different driving strategies on traffic risks. The structure of the causal graph can be expressed by the following equations:

$$X \rightarrow T \rightarrow Y. \quad (16)$$

For example, covariate X can be the speed and acceleration of surrounding vehicles, which can affect the handling variable T . T may be the driving decision to change lanes, and changing lanes may affect the outcome variable Y , which may be collision risk. So, establishing a causal relationship diagram can clearly show the causal relationships between variables. The data simulated by our HighwayEnv software is saved in CSV format, recording traffic information such as vehicle position, speed, acceleration, etc. These are all sources of causal reasoning data.

Next, we need the intervention operation $do(T)$ to analyze the causal effect of the treatment variable T on the outcome Y . In other words, we analyze the impact on Y under different T conditions by setting different T . For example, if in a traffic scene, we can make three driving decisions: changing lanes, accelerating, and decelerating, then we will

analyze the traffic risk impact of each of these three decisions one by one. So, this formula can be expressed as follows:

$$\mathbb{E}[Y | do(T)] = \sum_X \prod_{i=1}^n \mathbb{E}[Y | T, X, r_1, r_2, \dots, r_i] P(X | r_1, r_2, \dots, r_i), \quad (17)$$

where $\mathbb{E}[Y | do(T)]$ represents the expected value of Y after performing the intervention $do(T)$. $\mathbb{E}[Y | T, X, r_1, r_2, \dots, r_i]$ represents the expected value of Y given T , X , and the conditions of each step in the reasoning chain. $P(X | r_1, r_2, \dots, r_i)$: Represents the distribution of the covariate X after reasoning steps $r_1, r_2, \dots, r_i$.

In the causal inference model for traffic risk, the ATE (Average Treatment Effect) quantifies the impact of driving decisions (such as acceleration, deceleration, lane changes, etc.) on traffic risk. The formula for ATE is as follows [45]:

$$ATE = \mathbb{E}[Y(T=1)] - \mathbb{E}[Y(T=0)], \quad (18)$$

where $T = 1$ means a certain action is taken, while $T = 0$ means no action is taken, that is, maintaining the existing action. $Y(T=1)$ denotes the value of the outcome variable Y under the condition where $T = 1$ (action is taken), and similarly, $Y(T=0)$ denotes the value of Y when no action is taken. Thus, $\mathbb{E}[Y(T=1)]$ represents the expected traffic risk when a certain driving action is taken, and $\mathbb{E}[Y(T=0)]$ represents the expected traffic risk when no action is taken.

The backdoor criterion [46] is often used in the transportation field. By combining causal diagrams and backdoor criteria, we can effectively identify and adjust variables to estimate causal effects. In a given directed acyclic graph G , there is a pair of ordered variables (X, Y) , if the set of variables Z satisfies the following condition: the first condition is that none of the nodes in Z are descendants of X . The second condition is that Z blocks all paths between X and Y that point to X (in other words, backdoor paths). Then it is said that Z satisfies the backdoor criterion regarding (X, Y) . For example, vehicle A affects the user's lane change decision (T) (acceleration of vehicles behind may hinder lane change). Directly affecting the safety after lane change (Y), which complies with the backdoor criterion and requires adjusting variables. Vehicle B only indirectly affects safety (Y) by causing lane change (T), and does not directly affect Y , so there is no need to adjust variables.

Although the CI-COT method improves the interpretability of reasoning, it also relies on high-quality causal model construction. For example, the accuracy and completeness of handling causal relationships such as variables and covariates can significantly affect the reliability of reasoning while also increasing the burden of system development and maintenance. In the future, we will consider modularizing the causal relationship modeling process, developing automated tools to assist in the construction of causal models, reducing reliance on large amounts of annotated data and professional knowledge, and improving the maintainability of the system.

Finally, we integrate causal inference into the inference process of CoT and write it in pseudocode form, as shown in Algorithm 1. CI-CoT can help the TrafficRiskGPT model capture information related to traffic risks at every step of the inference process, and ATE can also quantify the impact of driving decisions on traffic safety, further making the output of the TrafficRiskGPT model more reasonable and reliable.

Lemma 1. We can adjust the covariate X to infer the expected value of $P(Y | T, X)$ based on the observed $P(Y | T, X)$. In other words, we can infer the causal relationship of an action in the current traffic scene. The causal effect of X and Y can be identified and is given by the formula:

$$\mathbb{E}[Y | do(T)] = \sum_X \mathbb{E}[Y | T, X] P(X). \quad (19)$$

Proof. [47] has proved this formula, so we will not repeat the proof here.

Algorithm 1 CI-COT

```

1: Input: Treatment variables  $T$ , outcome variables  $Y$ , covariates  $X$ , problem  $q$ 
2: Output: Rationale  $r_n$ , decision  $d$ 
3: Construct the input  $X_0$ , considering both  $X$  and problem  $q$ 
4: Perform Chain of Thought reasoning to derive intermediate rationales  $r_1, r_2, \dots, r_n$ 
5:  $r \leftarrow g(X, q)$ 
6: Concatenate  $r_n$  to  $X$ , making  $X' \leftarrow [X; r_n]$ 
7: Causal Inference: Apply Causal Inference technique
8: Perform ATE analysis for treatment effect on outcome:
9:   Calculate the expected outcome under treatment  $T = 1$ 
10:  Calculate the expected outcome under no treatment  $T = 0$ 
11: Compute the ATE:  $ATE = \mathbb{E}[Y | T = 1] - \mathbb{E}[Y | T = 0]$ 
12: Integrate ATE into Chain of Thought by updating the rationale  $r_n$  with causal information
13:  $r_n \leftarrow r_n + F_{causal}(ATE)$ 
14: Construct new input  $X'' = \{X', X_0, q\}$ 
15: Perform final Chain of Thought reasoning step with the updated input
16: Compute the final decision based on ATE and updated rationale  $r_n$ 
17:  $d \leftarrow F_{decision}(ATE, r_n, q)$ 
18: Output the final decision and causal reasoning analysis:
19: Return:  $r_n$  and  $d$ 

```

Lemma 2. Formula (16) has transitivity.

Proof. We first start from the causal inference formula (18) and introduce the marginalization of X , as follows:

$$\mathbb{E}[Y | do(T)] = \sum_X \mathbb{E}[Y | T, X] P(X). \quad (20)$$

Next, we introduce the CoT reasoning chain $r_1, r_2, \dots, r_n$ into causal inference, yielding:

$$\mathbb{E}[Y | T, X] \rightarrow \mathbb{E}[Y | T, X, r_1, r_2, \dots, r_n]. \quad (21)$$

Then, the distribution $P(X)$ expands to the conditional distribution that evolves along the reasoning chain:

$$P(X) \rightarrow P(X | r_1, r_2, \dots, r_i). \quad (22)$$

Finally, combining the updates of X , Y , and each reasoning step r_i along the chain, we have formula (16).

Lemma 3. The joint probability of each step in the formula $\prod_{i=1}^n p(r_i | do(r_{i-1}, \dots, r_1, q))$ can be recursively calculated.

Proof. According to the chain rule of causal inference, the impact of the intervention $do(r_1, r_2, \dots, r_{i-1})$ on the current reasoning step r_i can be described by the conditional probability:

$$p(r_i | do(r_{i-1}, r_{i-2}, \dots, r_1)) = p(r_i | r_{i-1}, r_{i-2}, \dots, r_1). \quad (23)$$

For step 1, r_1 :

$$p(r_1 | q), \quad (24)$$

For step 2, r_2 :

$$p(r_2 | do(r_1)) = p(r_2 | r_1). \quad (25)$$

For step 3, r_3 :

$$p(r_3 | do(r_2, r_1)) = p(r_3 | r_2, r_1). \quad (26)$$

For step i :

$$(r_i | do(r_{i-1}, r_{i-2}, \dots, r_1)) = p(r_i | r_{i-1}, r_{i-2}, \dots, r_1). \quad (27)$$

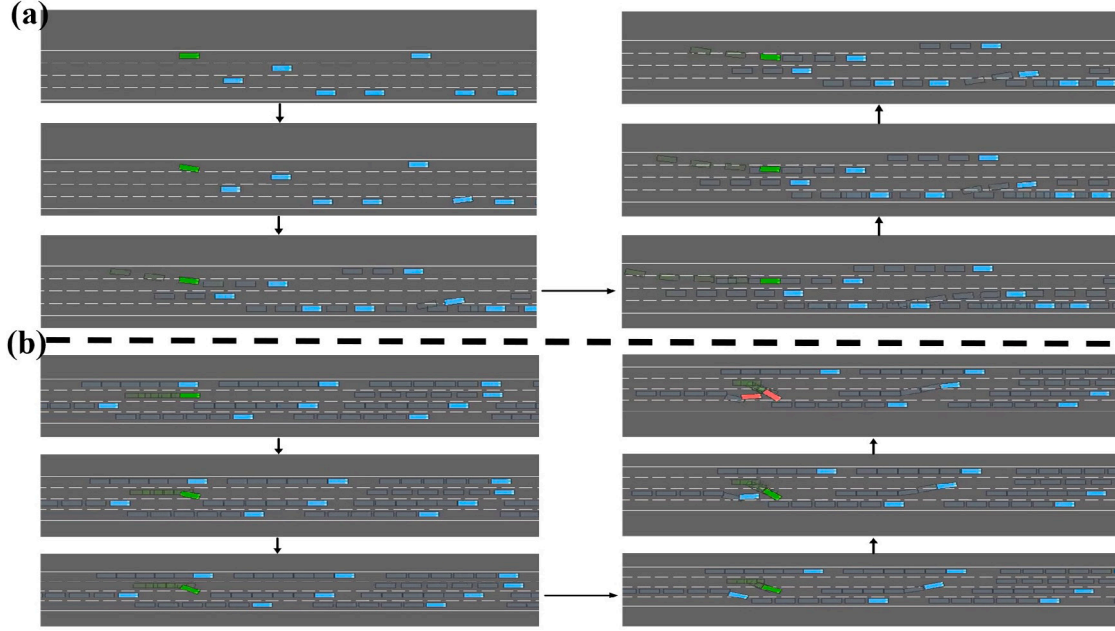


Fig. 5. Visualize the experimental results of successful and failed lane-changing traffic scenes, with green markings indicating ego vehicles, blue markings indicating other vehicles, and red markings indicating collision. (a) representing successful cases, (b) representing failed cases.

Combining the conditional probabilities at each step, the joint probability for the entire reasoning chain is:

$$p(r_1, r_2, \dots, r_n | do(q)) = p(r_1 | q) \prod_{i=2}^n p(r_i | r_{i-1}, \dots, r_1). \quad (28)$$

This formula indicates that each step in reasoning can be recursively calculated.

5. Experiment

In this section, we validate the performance of the TrafficRiskGPT model in traffic risk assessment and evaluate the effectiveness of the CI-CoT method.

5.1. Experimental configuration

The hardware environment includes a computer equipped with an 80-core Intel Xeon Gold 6133 CPU @ 2.50 GHz, as well as four NVIDIA RTX A6000 graphics cards, each with 48 GB of memory. The total memory is 384 GB, and the storage capacity is 11T in total. In terms of software environment, the system runs on the Ubuntu 20.04.1 LTS operating system, with a Python version of 3.10.0. Important Python library versions include NumPy 1.21.6, pandas 2.0.3, torch 1.13.1+cu116, TensorFlow 2.10.0, and transformers 4.33.0.

5.2. Knowledge base construction

To improve the safety and dependability of autonomous driving systems further, we developed a new dataset based on traffic scene risk, named HighRoadRisk. The HighRoadRisk dataset is based on the HighwayEnv simulation environment, generating different types of road traffic scenes. The default configuration of HighwayEnv is mainly designed for highway scenarios, which have certain limitations. For example, it is difficult to simulate traffic scenes where pedestrians cross the road, and there are certain differences between the simulated data of HighwayEnv and real-world data. This dataset contains information such as the position, speed, and acceleration of vehicles, as well as available action options such as acceleration, deceleration, maintaining status quo, left lane change, and right lane change. Through this

simulation environment, we constructed a series of specific video cases, with a training set of 780 cases and a testing set of 160 cases, covering different measures taken by vehicles to prevent traffic accidents in different scenes. Through these video cases, we can learn relevant knowledge strategies, such as what strategies vehicles can adopt to successfully avoid accidents or under what circumstances they might fail, thereby further improving vehicular safety.

In the dataset demonstration section, we selected two cases: One is a successful lane-changing case, as shown in Fig. 5(a), demonstrating that under specific conditions, adopting the correct strategy can smoothly complete the lane change without affecting the traffic of other vehicles. the second scene depicts a collision occurring during the lane change, as illustrated in Fig. 5(b). We can analyze the causes of the collision and possible avoidance strategies to provide an important data basis for model improvement.

The HighRoadRisk dataset also includes more traffic events, as shown in Fig. 6. Cases such as the sudden cut-in scene, vehicle coalescence scene, and the sudden breaking scene are added to enrich the knowledge base and cover as many traffic scenes as possible, thereby reducing the difference between simulated data and real-world data and improving the model's generalization ability. Despite the fact that we have added some sudden emergency cases to our knowledge base, there are still shortcomings in our emergency response mechanism when our system encounters vehicle failures or multiple failures. For example, when a vehicle experiences both sensor and power system failures simultaneously, how can the system comprehensively consider various fault factors to develop the optimal emergency plan. Currently, there are certain limitations to such extreme scenes.

We also tested the retrieval time comparison of different methods in the knowledge base, selecting FlatL2, IVFFlat, PQ (Product Quantization) and HNSW (Hierarchical Navigable Small World graphs) for comparison.

As shown in Fig. 7, we can see that the retrieval time of the FlatL2 method increases almost linearly with the increase of the number of vectors. The reason is that the FlatL2 method uses brute force search to calculate the distance of all vectors, resulting in lower efficiency. The IVFFlat method utilizes inverted lists and clustering methods to reduce the number of retrieval vectors, resulting in much higher retrieval efficiency than the FlatL2 method. The PQ method reduces computation

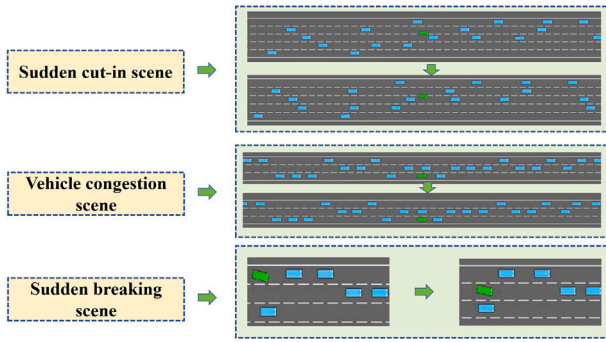


Fig. 6. More traffic case studies to showcase.

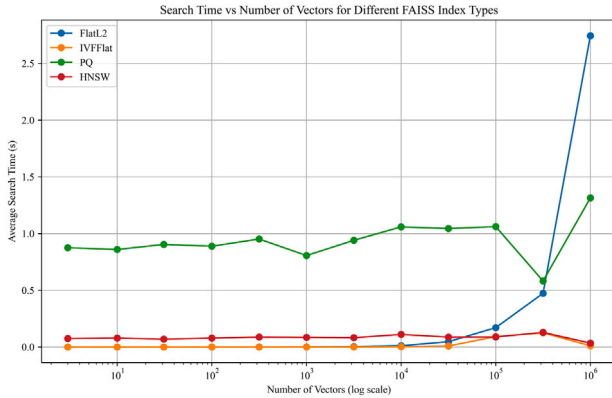


Fig. 7. Comparison of different methods in search time and vector quantity.

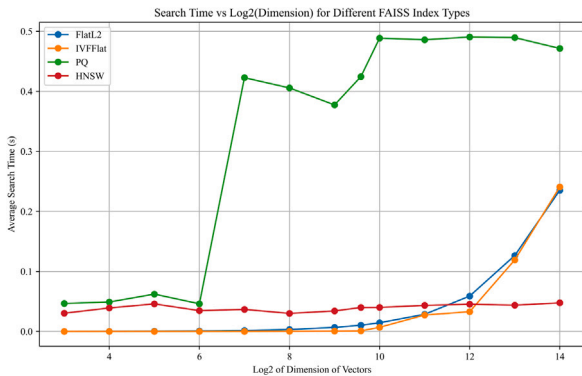


Fig. 8. Comparison of different methods in search time and vector dimension.

and storage through quantization, but still has some approximation error. The HNSW method is an approximate nearest neighbor search method based on graph structure, which can quickly converge and maintain low retrieval time. In this experiment, the IVFFlat method is similar.

As shown in Fig. 8, we also tested the retrieval time variation under different vector dimensions with a fixed number of vectors. The FlatL2 method significantly increases with the increase of dimensions, especially in high dimensions, showing poor scalability. IVFFlat method. It performs well in low dimensions, but the retrieval time gradually increases in high dimensions, indicating a decrease in efficiency in high dimensions. As the dimension of PQ method increases, the retrieval time also keeps increasing. The HNSW method maintains relatively low retrieval time in all dimensions, especially outstanding performance in high-dimensional data, and can balance computational efficiency well.

Table 1

Simulation collision rate and experimental completion rate of different large models.

Model	Collision rate	Completion rate
IDM	0.00%	20.4%
MOBIL	0.00%	15.3%
DQN	3.75%	50%
Baichuan2-13B	19.61%	10%
ChatGPT3.5	5.51%	40%
GPT4o-mini	2.05%	70%
TrafficRiskGPT	1.96%	70%

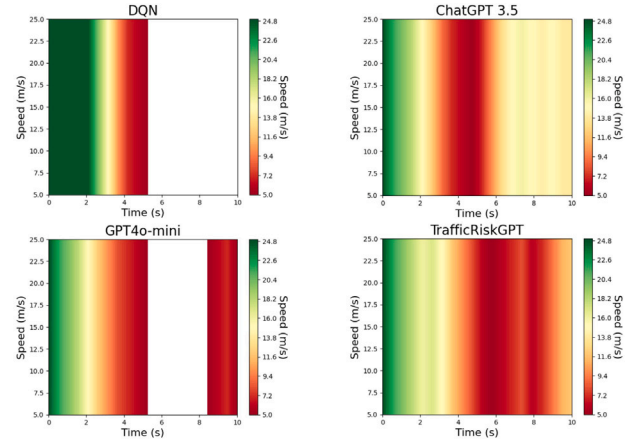


Fig. 9. Comparison chart of speed changes.

5.3. Various model simulation tests

We compared the collision rate and driving speed of traditional models DQN and LLM, as shown in Table 1. We selected the Intelligent Driver Model (IDM) [48] and the Minimizing Overall Braking Induced by Lane Changes (MOBIL) [49] from Ref. [50] as baselines, both of which are rule-based methods with collision rates of 0 and completion rates within 20%. We found that under the same conditions, the collision rate of our constructed TrafficRiskGPT was significantly lower than other models. Compared with the traditional model DQN, the collision rate was reduced by 47.73%, and compared with ChatGPT3.5, the collision rate was reduced by 64.43%. Compared with the latest GPT4o-mini model, the collision rate has decreased by 7.3%. In addition, in terms of completion rate, GPT4o-mini and TrafficRiskGPT had the highest completion rate, while Baichuan2-13B had the slowest.

Next, we compared the average speed of our own vehicles in the same scene, as shown in Fig. 9. We found that DQN has the slowest average speed. Further demonstrating the driving speed of various vehicle models, we found that DQN experienced deceleration and the driving speed dropped to its lowest point, approaching zero. GPT4o-mini, like DQN, also adopts a conservative deceleration strategy to avoid collisions as much as possible. Then, after experiencing deceleration, ChatGPT 3.5 and TrafficRiskGPT gradually increased their speed, which is more in line with our daily driving behavior and proves the feasibility of TrafficRiskGPT.

5.4. vLLM engine enhancement mechanism

We employed the vLLM as an acceleration platform for large models to increase the output token speed of LLMs. We compared the token output speed and memory consumption of Baichuan2-13B and TrafficRiskGPT under one-shot conditions with and without the vLLM. As shown in Fig. 10, on the vLLM platform, the token output of Baichuan2-13B increased by 23.40 times, and memory consumption decreased by 59.18%. The token output of TrafficRiskGPT increased by 3.63 times, and memory consumption decreased by 30.16%.

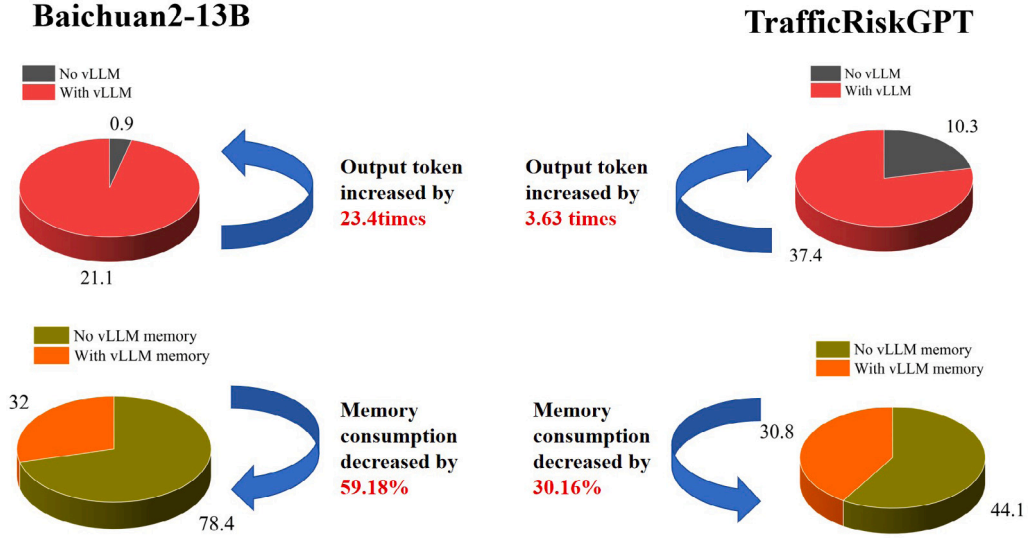


Fig. 10. vLLM Engine test.

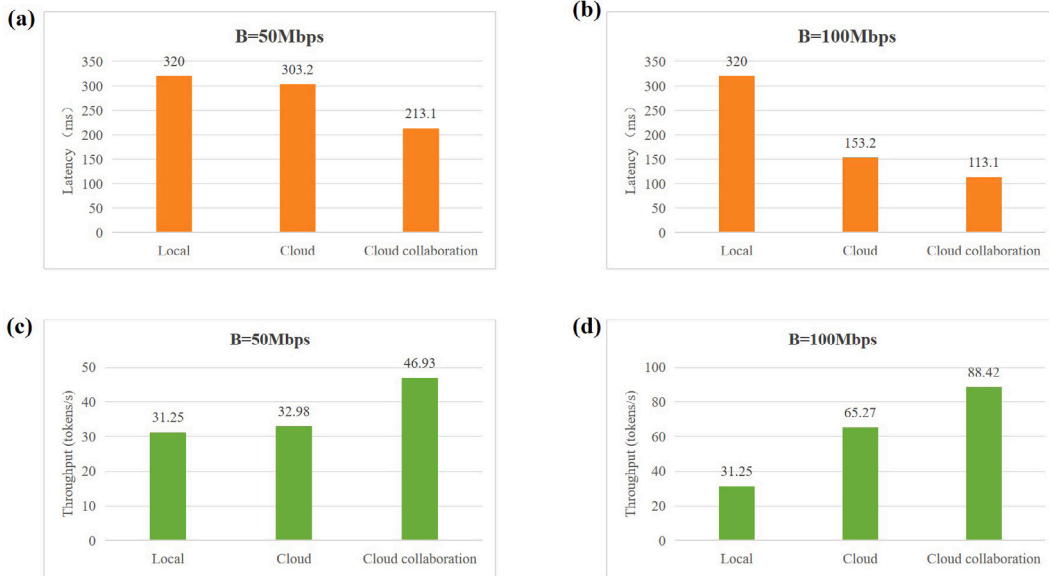


Fig. 11. Collaborative TrafficRiskGPT inference latency and throughput under different bandwidths.

We have set the bandwidth between the in car and cloud servers to two scenarios: 50 Mbps and 100 Mbps. As shown in Fig. 11, in terms of latency, we compared Fig. 11(a) and (b) and found that, in addition to fully local inference, both fully cloud based and cloud based collaborative inference showed a continuous decrease in latency with increasing bandwidth, with the most significant decrease observed in cloud based collaborative inference. In terms of throughput, comparing Fig. 11(c) and (d), we can find that in addition to fully local inference, the throughput of fully cloud based and cloud based collaborative inference also increases with the increase of bandwidth, with the most significant increase in throughput for cloud based collaborative inference.

5.5. Assessment of traffic risk assessment capability for various models

In the current highway traffic scene, we refer to Ref. [50] and consider TTC, DRAC and DSS as indicators that have a significant impact on traffic risk. Therefore, we assign them relatively high weights.

The weights W_T , W_{DR} , W_{DS} and W_{DC} are set to 0.3, 0.3, 0.3 and 0.1. We used the driving scores of the IDM model and MOBIL model as the baseline, with the IDM model scoring 17.3 and the MOBIL model scoring 11.7. We found that LLM's driving score was higher than rule-based methods. As shown in Fig. 12, we further analyzed that the TTC score of DQN was 53.09, the DRAC score was 58.71, the DSS score was 40.44, the DC score was 22.77, and the DS score was 41.88. DQN achieved a relatively good risk assessment score, but considering that the average speed of driving decisions based on DQN is very slow and does not match the actual driving speed.

Subsequently, we compared the TTC, DRAC, DSS, DC, and DS assessment scores of Baichuan2-13B, ChatGPT3.5, Gpt4o-mini and TraficRiskGPT. As shown in Fig. 12, we found that the various traffic risk assessment indicators of TrafficRiskGPT and GPT4o-mini are not much different. In addition, our TrafficRiskGPT's DS metric is 39.40% higher than the DQN method and 31.07% higher than ChatGPT 3.5. In the Risk

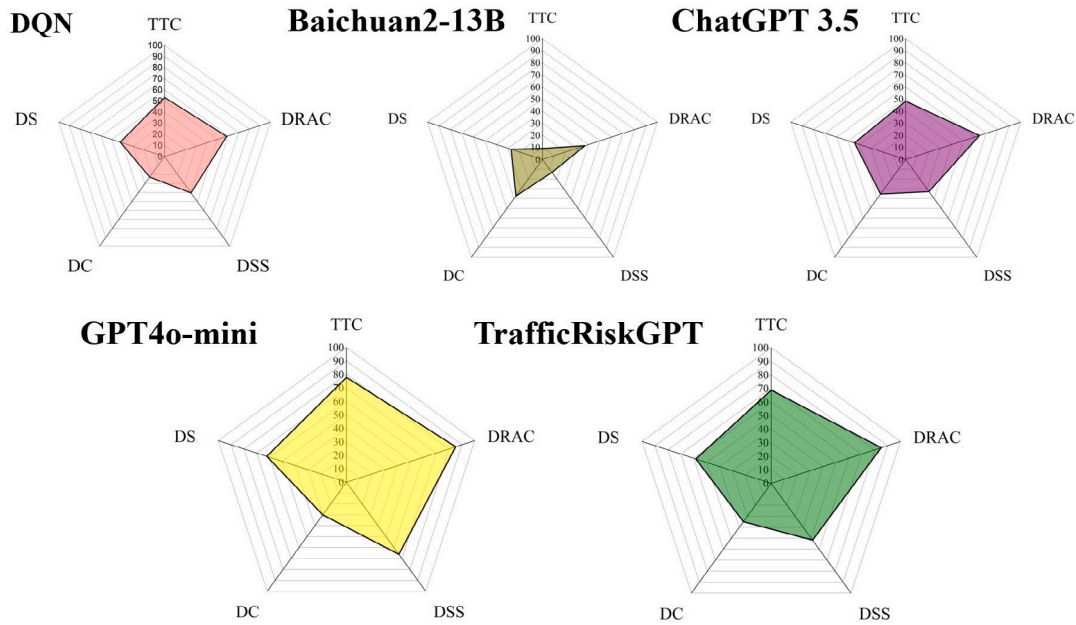


Fig. 12. Comparison of traffic risk assessment indicators.

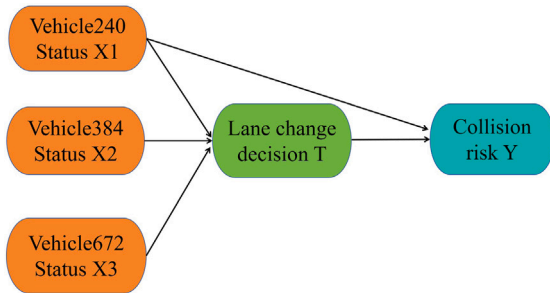


Fig. 13. Causal graph.

Score, such as TTC score, our TrafficRiskGPT's TTC metric is 42.69% higher than ChatGPT 3.5.

5.6. CI-CoT test

Firstly, we constructed a causal diagram of the traffic scene, as shown in Fig. 13. Then use the backdoor criterion to identify causal variables. Next, we conducted an analysis of CI-CoT. We analyzed one of the traffic scenes in the dataset, as shown in Fig. 14. We presented the causal relationships between different variables in this traffic scene, such as speed, acceleration, distance from the preceding vehicle, ATE, and driving behavior, represented by color depth. Red represents positive causal relationships, blue represents negative causal relationships, and color depth represents the strength of causal effects. The darker the color, the stronger the causal relationship. As shown in Fig. 14(a), the distance from vehicle 240 has a negative impact on the risk of traffic accidents, indicating that increasing the distance can reduce the risk of accidents. The impact of driving behavior on accident risk is positively correlated, indicating that such behavior (such as acceleration) may increase the risk of traffic accidents. Through this visualization, we can intuitively see which variables have a significant impact on collision risk and driving behavior, which helps to make safer driving decisions.

Finally, we conducted a comparative analysis between CoT and CI-CoT, as shown in Fig. 15. The final decision was to slow down, and we found that the output of CoT mainly provided a series of logical reasoning operations without clear causal relationships. As shown in Fig. 16, CI-CoT incorporates causal analysis in the gradual inference process and can evaluate the final traffic causal effects of each decision, making the model output more accurate and interpretable.

6. Conclusion

We have designed a traffic scene risk assessment service system based on the TrafficRiskGPT model, aiming to provide reliable and trustworthy traffic scene evaluation services. Our constructed TrafficRiskGPT model is based on the LLaMA3-8B architecture and is fine-tuned using LoRA with traffic risk scene data. The system enhances the model's reliability through a knowledge base composed of scene descriptions, various driving tasks, and decision plans, and combined with the HNSW method to improve the retrieval efficiency of the knowledge base. Simultaneously, it introduces vLLM technology to improve the model's inference speed and uses risk assessment metrics such as TTC and DRAC to evaluate the model's performance. Finally, we designed a CI-CoT technical method to improve the TrafficRiskGPT model's performance in capturing risk factors in traffic scenarios, making the model's reasoning process closer to real traffic scenes and more transparent. The experimental results indicate that the collision rate of the TrafficRiskGPT model was reduced by 47.73% compared with that of the traditional model DQN, and reduced by 64.43% compared with that of ChatGPT3.5. After system reinforcement, the token output of TrafficRiskGPT increased by 3.62 times, and memory consumption decreased by 30.16%. The inference speed significantly increases while the irrelevant output significantly decreases, which illustrates the issues of model illusion and slow inference speed are alleviated. Due to the difference between the data simulated by HighwayEnv simulation software and the real world data, it may be difficult to cope with unknown risks when applied in the actual auto drive system. In future research, we can further deploy the model in the real scene to further improve the reliability of the model.

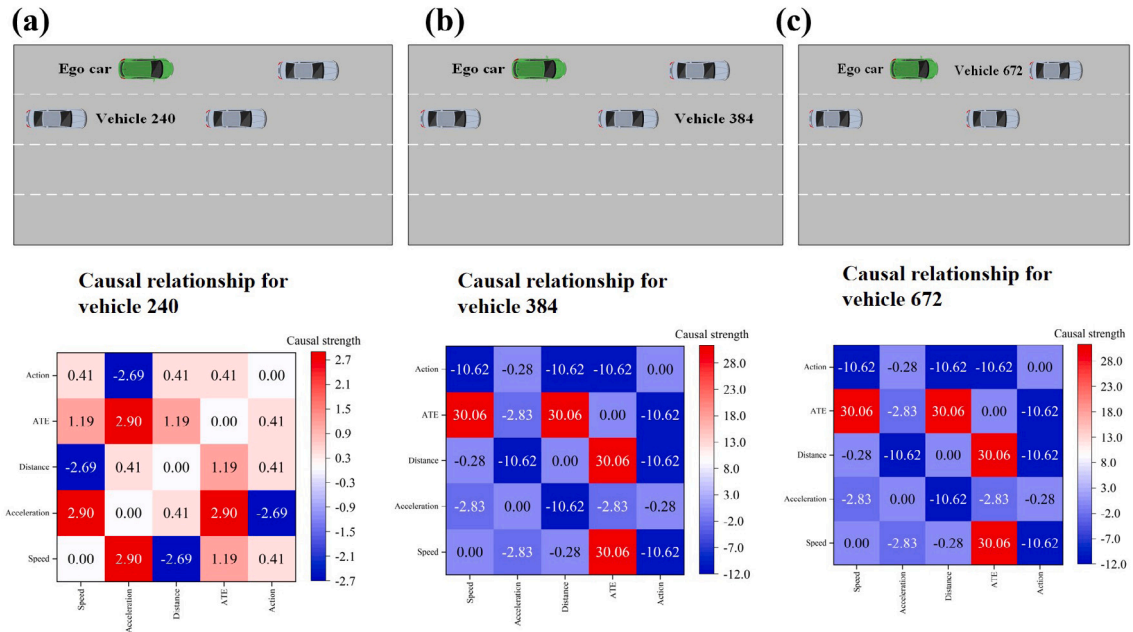


Fig. 14. Cause and effect diagram of traffic scene.

You are driving on a road with 4 lanes, and you are currently driving in the leftmost lane. Your current position is `(433.92, 0.00)`, speed is 15.80 m/s, acceleration is -1.27 m/s^2 , and lane position is 433.92 m. There are other vehicles driving around you, and below is their basic information:

- Vehicle `240` is driving on the lane to your right and is behind of you. The position of it is `(432.73, 4.75)`, speed is 12.90 m/s, acceleration is 1.42 m/s^2 , and lane position is 432.73 m.
- Vehicle `384` is driving on the lane to your right and is ahead of you. The position of it is `(463.98, 4.00)`, speed is 18.63 m/s, acceleration is -0.99 m/s^2 , and lane position is 463.98 m.
- Vehicle `672` is driving on the same lane as you and is ahead of you. The position of it is `(507.47, 0.02)`, speed is 20.28 m/s, acceleration is -2.18 m/s^2 , and lane position is 507.47 m.



In this scenario, I am currently in the leftmost lane and there is a vehicle ahead of me in the same lane (Vehicle 672) that is moving slightly slower than I am. Additionally, there is a vehicle to my right (Vehicle 384) that is also ahead of me but moving even slower.

Given that my speed is higher than both vehicles ahead, I need to consider my options carefully to avoid a potential collision. If I continue at my current speed, I will eventually close the gap with Vehicle 672, which could lead to a rear-end collision if it continues to decelerate.

I have the option to decelerate, which would help maintain a safe distance from Vehicle 672. Alternatively, I could consider changing lanes to the right, but that would put me in a position where I might still be close to Vehicle 384, which is also decelerating.

To prioritize safety and avoid collisions, the best action is to decelerate and create more space between myself and the vehicle ahead.

Fig. 15. Testing scene with CoT technology.

Same scene description as CoT.



In the current scenario, I am driving in the leftmost lane at a speed of 15.80 m/s with a slight deceleration. There are two vehicles in the lane to my right: one behind me (Vehicle 240) and one ahead of me (Vehicle 384). Additionally, there is a vehicle ahead of me in my current lane (Vehicle 672) that is traveling faster than I am, which could lead to a potential collision if I do not adjust my speed or position.

Given that Vehicle 672 is ahead of me and moving at 20.28 m/s, I need to consider my options carefully. If I continue to maintain my current speed, I will eventually close the gap with Vehicle 672, which is decelerating. This could lead to a situation where I am too close to the vehicle ahead, increasing the risk of a rear-end collision.

Turning right to change lanes (Action_id: 2) could be a viable option, but I need to ensure that Vehicle 240, which is behind me in the right lane, is not too close to me. Vehicle 240 is currently traveling at 12.90 m/s and is behind me, which means it has some distance to cover before it reaches my position. However, I must also consider the speed of Vehicle 384 ahead in the right lane, which is moving faster than me at 18.63 m/s. If I were to change lanes, I could potentially find myself in a situation where I am sandwiched between two faster-moving vehicles.

Decelerating (Action_id: 4) would allow me to create more space between myself and Vehicle 672, reducing the risk of a collision. This action would also give me more time to assess the situation and decide whether to remain in my lane or change lanes later.

Acceleration (Action_id: 3) is not advisable at this moment, as it would only increase my speed and potentially worsen the situation with Vehicle 672 ahead of me.

Given these considerations, the safest action is to decelerate, allowing for more space and time to react to the vehicles around me.

Fig. 16. Testing scene with CI-CoT technology.

CRediT authorship contribution statement

Wuchang Zhong: Writing – review & editing, Validation, Methodology. **Jinglin Huang:** Software. **Maoqiang Wu:** Supervision, Funding acquisition. **Weinan Luo:** Data curation. **Rong Yu:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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